Analysis of the effect of cognitive load on MPLs choices

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1 Motivation: no visible effect of cognitive load on deviations from expected value

The analysis of cognitive load starts from the fact that no consistent effect emerges when deviations are plotted by tasks and differentiate between the cognitive load treatment and the no-cognitive load treatment (Figure 1).

2 Investigation of the effect of cognitive load on choices using the correlation of different variables with the deviations

2.1 Three correlations/correlations to assess

One might hypothesize that cognitive load has an impact only in the first part of the experiment (cog load on span task) and not on the main part with the choices. To investigate this hypothesis, I follow three paths. i) I test for the absence of a treatment effect by averaging all deviations across tasks. ii) I also examine the correlation between performance on the cognitive load task (span accuracy) and deviations. If the cognitive load manipulation works there should be a positive correlation between the performance on the cognitive load task (accuracy) and the absolute deviations: better remembering the digits would have no impact on choices. iii) I estimate the effect of sensitivity to cognitive load on deviations. Sensitivity quantifies how much participants memory is impacted by the cognitive load it encounters. Highly sensitive participants should exhibit stronger effects than those with low sensitivity. I analyze these questions in section in 2.2.

One may also suspect that the cognitive load is effective only with a time limit so that participants don't transfer the digits in long term memory before doing the choices. Thus we would observe an interaction effect between time limit and i) cognitive load, ii) accuracy and iii) cognitive load sensitivity. I analyze these questions in section in 2.3.

I analyze the effects on both absolute and normalized deviations. If cognitive load has an impact, it should increase the absolute deviation because of increased noise in the data. Given past research paper [Choi et al., 2022] it should also translate into higher normalized deviation.

2.2 Correlation plots

I begin with the analysis of the figures 2 and 3 (subject level). I analyse sensitivity to cognitive load (named "impactCogLoad" on the graphs), accuracy, and I also add time spent on the choices and maximum span.

1. The first line is linked to the third (iii) path I have outlined: the impact of cognitive load sensitivity on deviations. On the first line, only the plot in the cognitive load treatment is meaningful, as only participants in the cognitive load treatment have a cognitive load, and as the difference between their performance on the source and on the target spans is very dissimilar: participants in the no cognitive load treatment have a length one source span, participants in the cognitive load treatment has a maximum length source span. Analysis of this central plot reveals that the degree to which participants are affected by cognitive load does not correspond to an effect on their deviation levels, both for normalized (Figure 2) and absolute (Figure 3) deviations. At most, the effect is negative. This supports the hypothesis that the cognitive load

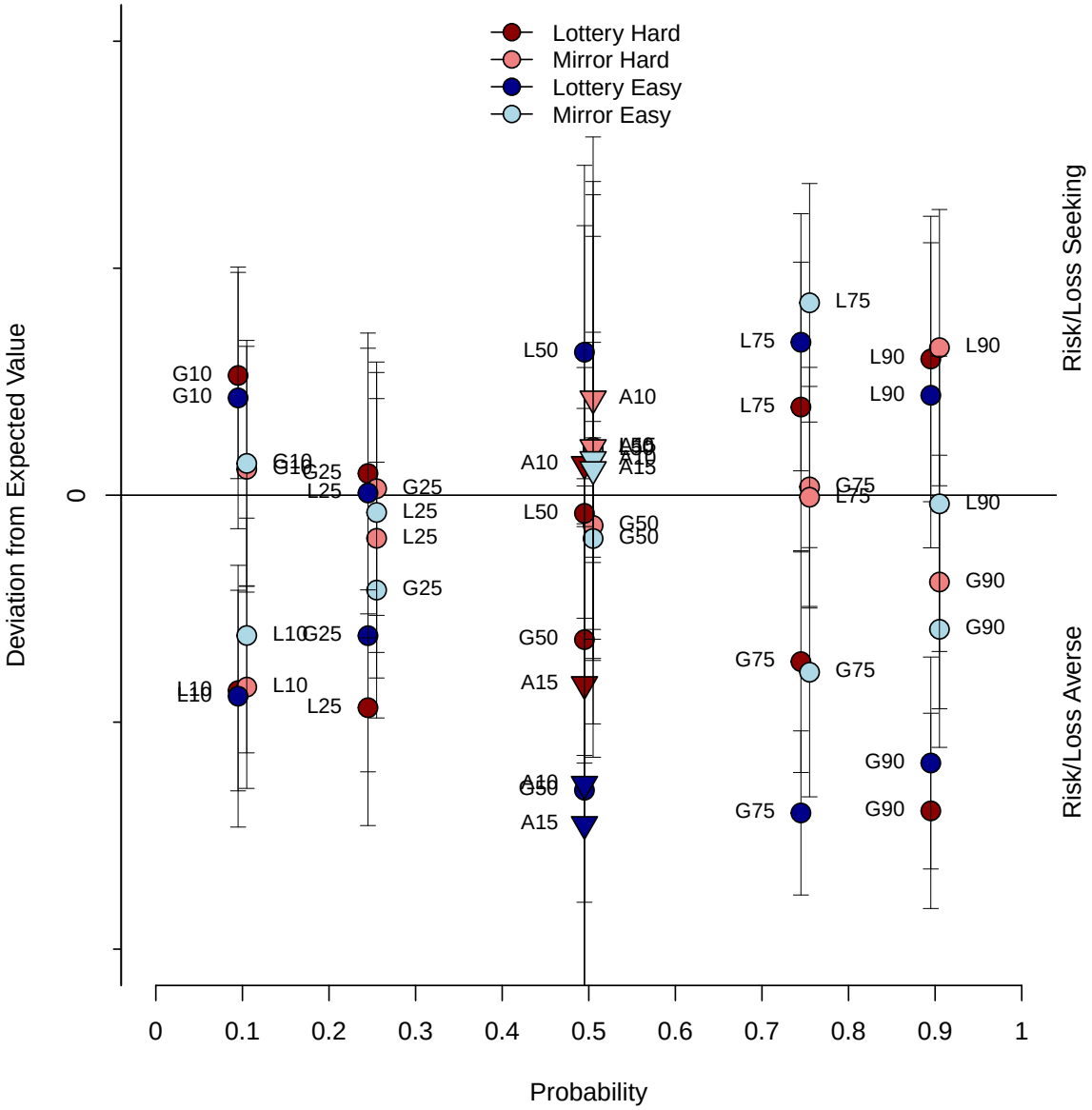

Figure 1: For fourfold lotteries, the y-axis measures the difference between subjects' certainty/simplicity equivalent and expected value, as stated in the axis label. The x-axis is the probability of the nonzero payoff. For loss aversion tasks, the y-axis measures instead the difference between the certain/simple payoff and the expected value of the mixed lottery/mirror. Two–standard error bars are included for every task.

does not have an impact and may not be well calibrated, as it is not even correlated with absolute deviations.

2. The second line analyzes the second (ii) path I have outlined. We can see in the second row of Figure 2 and 3 that accuracy is not correlated with normalized and only slightly with absolute deviation. Participants better at remembering the span, who presumably allocate more effort on the span (source task) do not exhibit more deviation from expected value, and not more in the direction predicted by PT. This indicates that the cognitive load has no impact on the choices.

3. The time taken to make choices has a global negative effect, in particular on absolute deviations. It is in keeping with the idea that the more participants invest time and thus effort into a decision the less they deviate from the gain maximizing choice (EV). For normalized deviations (Figure 2), it has a negative effect on lotteries in both treatments but on mirrors the effect is mixed between treatments.

4. Finally, the maximum span achieved at the beginning of the experiment, is in the cognitive load treatment positively correlated with the normalized and absolute (see Figure 3) deviations. Because the maximum span of the participants was used as the length of the span of the cognitive load, the positive correlation in the cognitive load treatment supports the hypothesis that the cognitive load exagerates the deviations in the direction predicted by prospect theory. In the control treatment, the maximum span is as awaited negatively correlated with absolute deviations as it is also impacted by the motivation and cognitive capacities of the participants.

In sum, the correlations presented here are non-conclusive: supporting the existence of an effect of cognitive load on deviations, the length of the span seems to increase the deviations. Supporting the idea that cognitive load has no impact, neither sensitivity nor accuracy correlated correlate with the deviations in the cognitive load treatment. The overall negative correlation between the time spent on choices and the deviations indicates that the deviations are sensitive to cognitive manipulations.

2.3 Analysis of the mediation effect of time

2.3.1 Correlations with median split

Reanalysis of the correlations suggests a mediating effect of time on cognitive load. Specifically, between participants who spend more time and those who spend less time the correlation between sensitivity to cognitive load and deviations is reversed. For both normalized (Figures 4 and 5) and absolute (Figures 6 and 7) deviations we observe a reversal of the effect of the sensitivity to cognitive load. This supports the idea that cognitive load does have an impact but only with a limited amount of time. Participants' accuracy on the source task (second row) displays a similar reversal pattern, thus also advocating for an effective cognitive load under time pressure. So path (ii) and (iii) do seem hold with a low time limit. These analyses should be taken with caution as there are only 11 participants in each treatment.

2.3.2 Verification of the interaction of time and sensitivity to cognitive load.

The interaction effect between time and cognitive load sensitivity can also be seen with linear models including an interaction term. I regress the deviation for each subject i, task type (e.g. G10) q and task status (mirror or lottery) s, on the mean time taken RT and the cognitive load sensitivity Cog.

$$dev_{iqs} = \alpha + \beta \cdot (\text{RT})_{iqs} + \gamma \cdot (\text{CogSen})_i + \varepsilon_{iqs}$$

The corresponding regression table is Table 1.

1. Cognitive load sensitivity does have a positive impact on both absolute and normalized deviations, once the effect of time is washed out. This advocates for the efficacy of the manipulation.

2. Time does only have an impact on absolute deviations. Thus the effect of time may be mostly due to noise, and the deviation in the direction of PT mediated by the sensitivity to cognitive load. This is not in keeping with [Choi et al., 2022], who finds a directional distorsion through probability weighting.

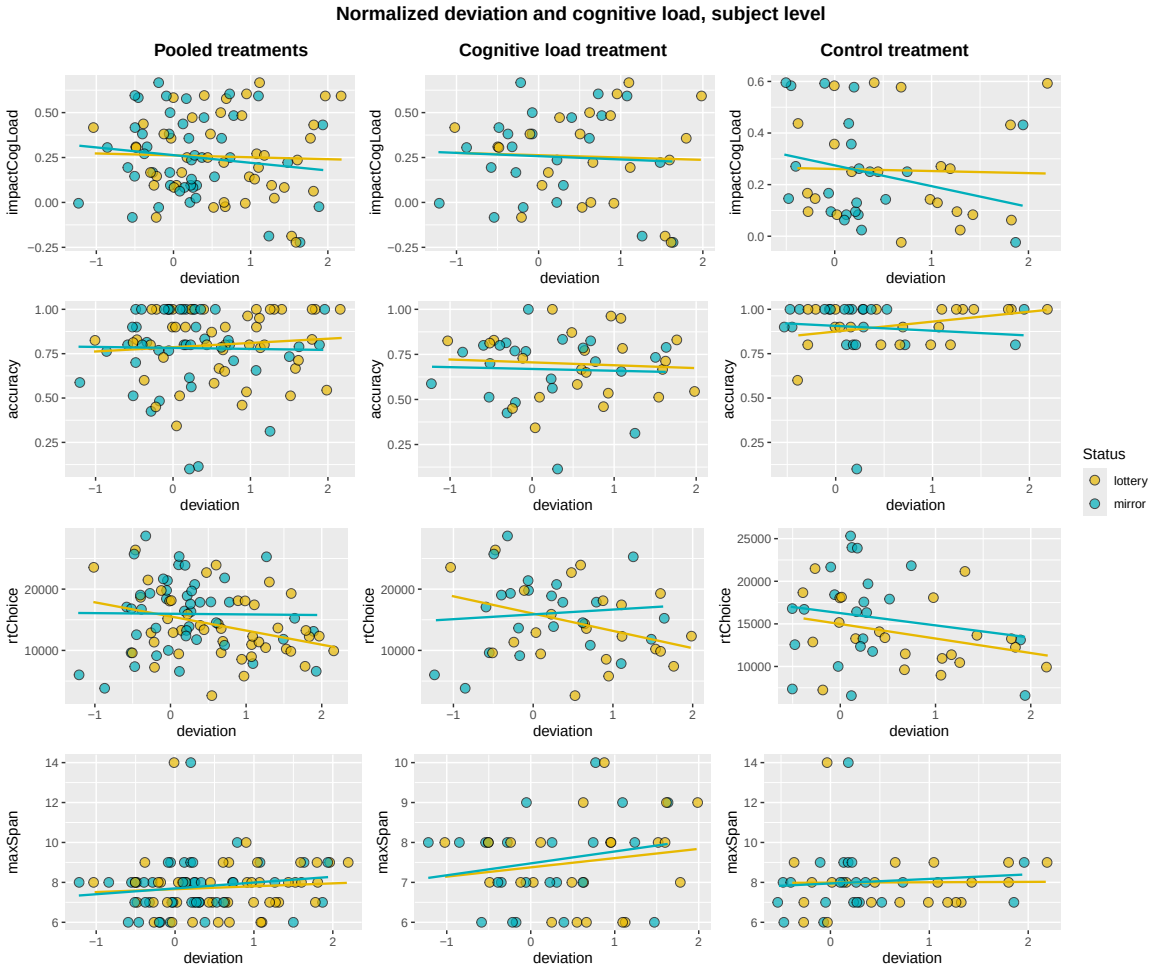

Figure 2: The x-axis shows the mean normalized deviation for each subject in the lottery (yellow) and mirror (blue) tasks. On the first row the y-axis represents the difference between the source task and the target tasks in the first part of the experiment. A very cognitive load sensitive subject would have a high positive difference. The y-axis of the second row represents the mean accuracy on the cognitive load task (span task) for the lotteries and mirrors tasks. On the third row the y-axis indicates the mean time that each participants has spent on the task (the MPLs). The last row's y-axis represents the maximum span of each subject. The lines are linear regression fits of the two variables, for lotteries and mirror choices. In total 45 subjects passed the comprehension questions, 23 are in the cognitive load treatment and 22 in the control treatment.

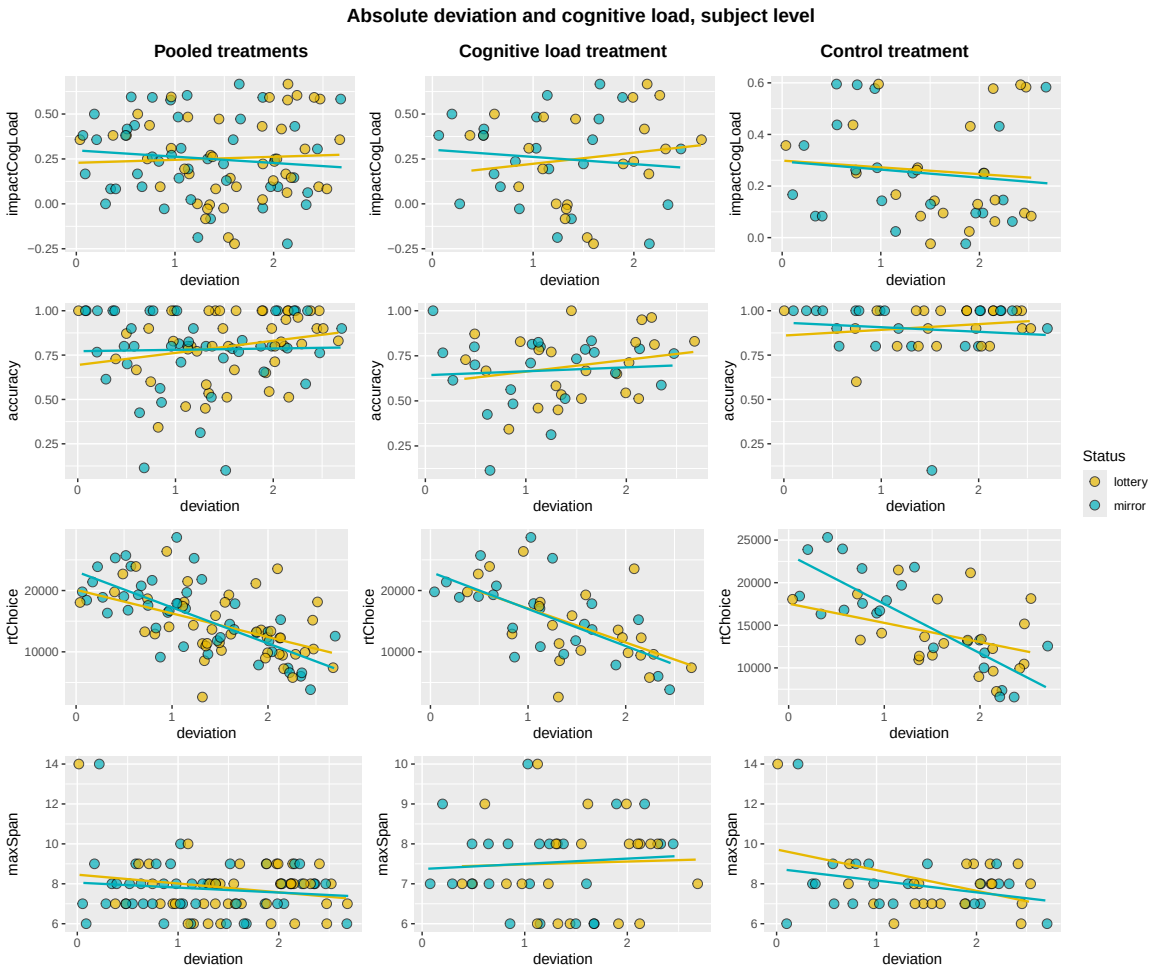

Figure 3: The x-axis represents the mean absolute deviation for each subject in the lottery (yellow) and mirror (blue) tasks. On the first row the y-axis represents the difference between the source task and the target tasks in the first part of the experiment. A very cognitive load sensitive subject would have a high positive difference. The y-axis of the second row represents the mean accuracy on the cognitive load task (span task) for the lotteries and mirrors tasks. On the third row the y-axis indicates the mean time that each participants has spent on the task (the MPLs). The last row's y-axis represents the maximum span of each subject. The lines are linear regression fits of the two variables, for lotteries and mirror choices. In total 45 subjects passed the comprehension questions, 23 are in the cognitive load treatment and 22 in the control treatment.

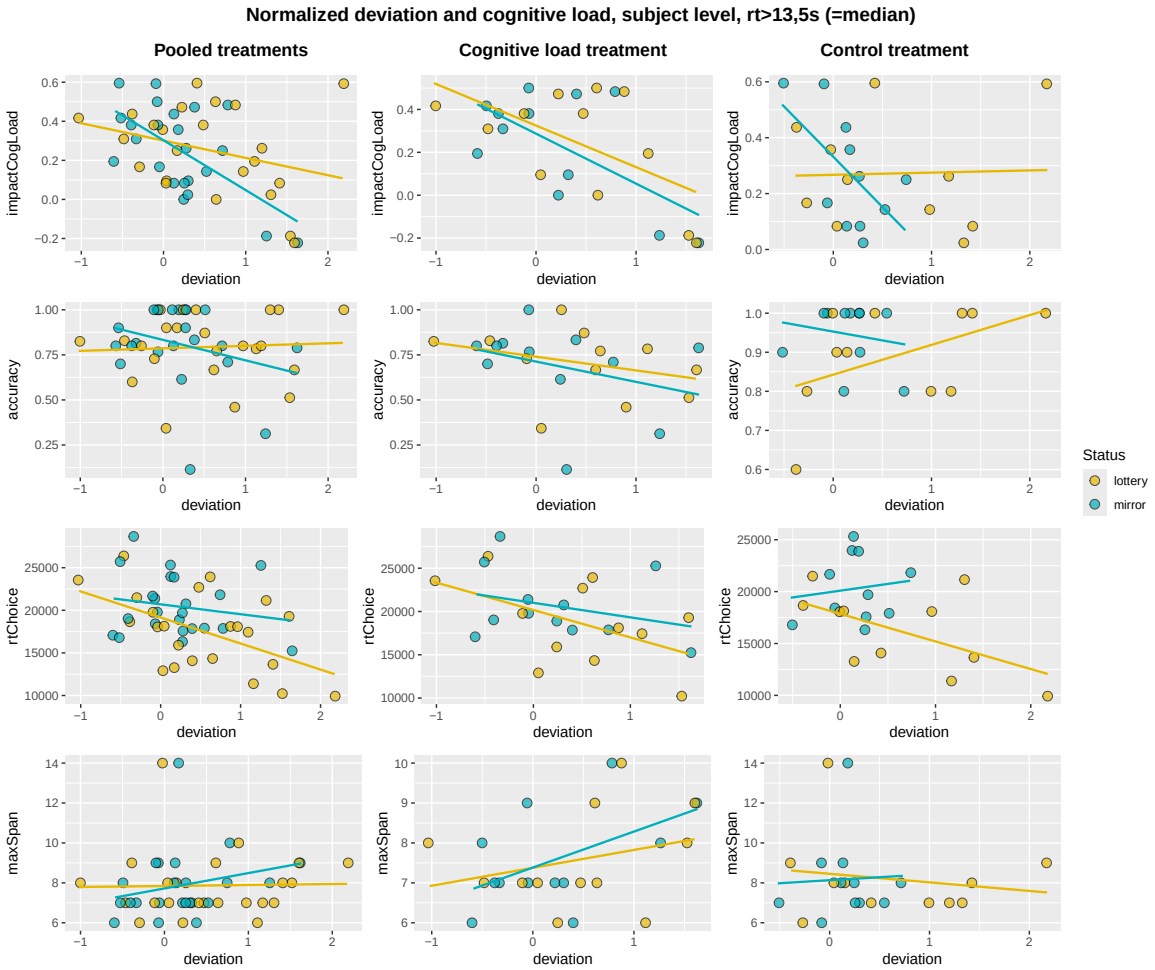

Figure 4: In this figure, all participants took on average more than 13,5 seconds to make their decisions on the choice tasks. The x-axis represents the mean absolute deviation for each subject in the lottery (yellow) and mirror (blue) tasks. On the first row the y-axis represents the difference between the source task and the target tasks in the first part of the experiment. A very cognitive load sensitive subject would have a high positive difference. The y-axis of the second row represents the mean accuracy on the cognitive load task (span task) for the lotteries and mirrors tasks. On the third row the y-axis indicates the mean time that each participants has spent on the task (the MPLs). The last row's y-axis represents the maximum span of each subject. The lines are linear regression fits of the two variables, for lotteries and mirror choices. In total 45 subjects passed the comprehension questions, 23 are in the cognitive load treatment and 22 in the control treatment.

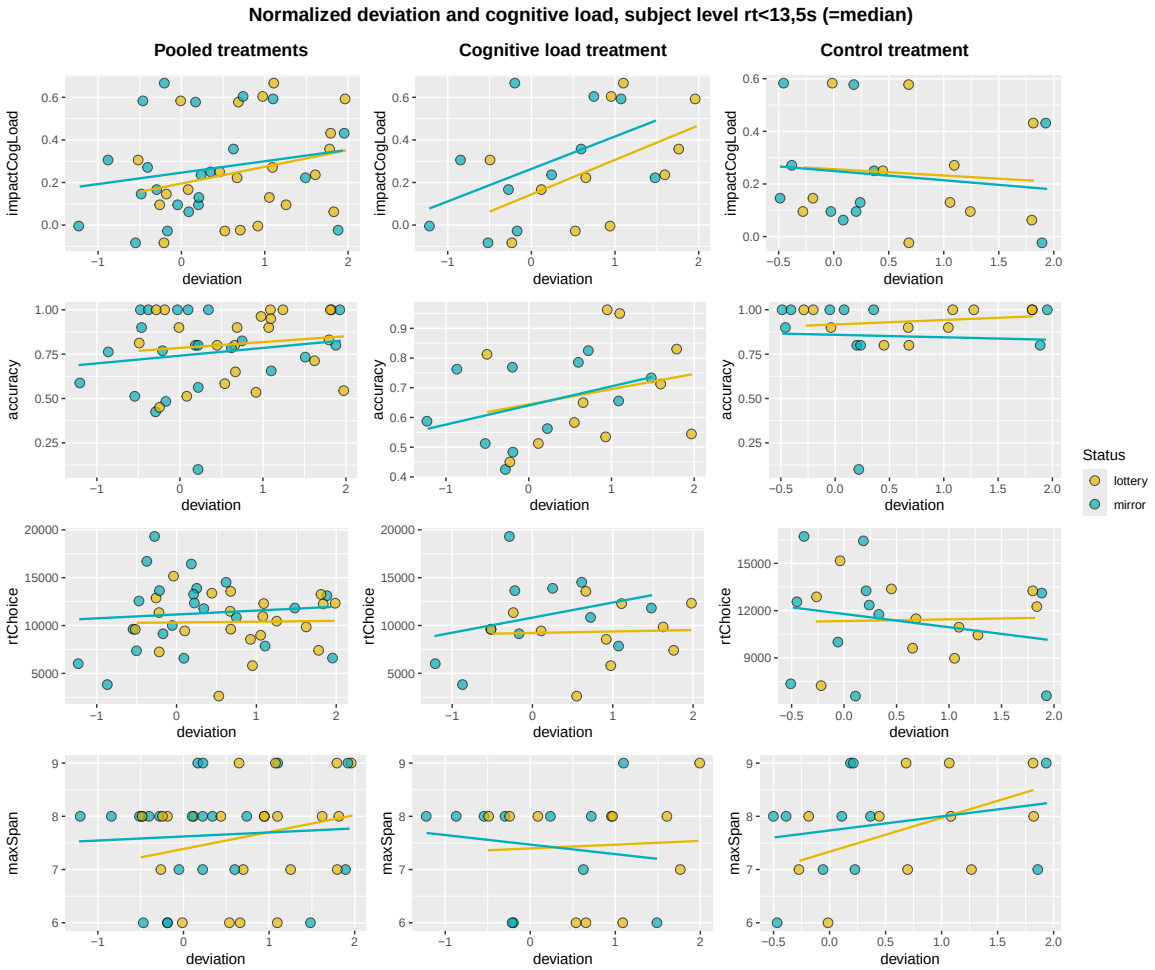

Figure 5: In this figure, all participants took on average less than 13,5 seconds to make their decisions on the choice tasks. The x-axis represents the mean absolute deviation for each subject in the lottery (yellow) and mirror (blue) tasks. On the first row the y-axis represents the difference between the source task and the target tasks in the first part of the experiment. A very cognitive load sensitive subject would have a high positive difference. The y-axis of the second row represents the mean accuracy on the cognitive load task (span task) for the lotteries and mirrors tasks. On the third row the y-axis indicates the mean time that each participants has spent on the task (the MPLs). The last row's y-axis represents the maximum span of each subject. The lines are linear regression fits of the two variables, for lotteries and mirror choices. In total 45 subjects passed the comprehension questions, 23 are in the cognitive load treatment and 22 in the control treatment.

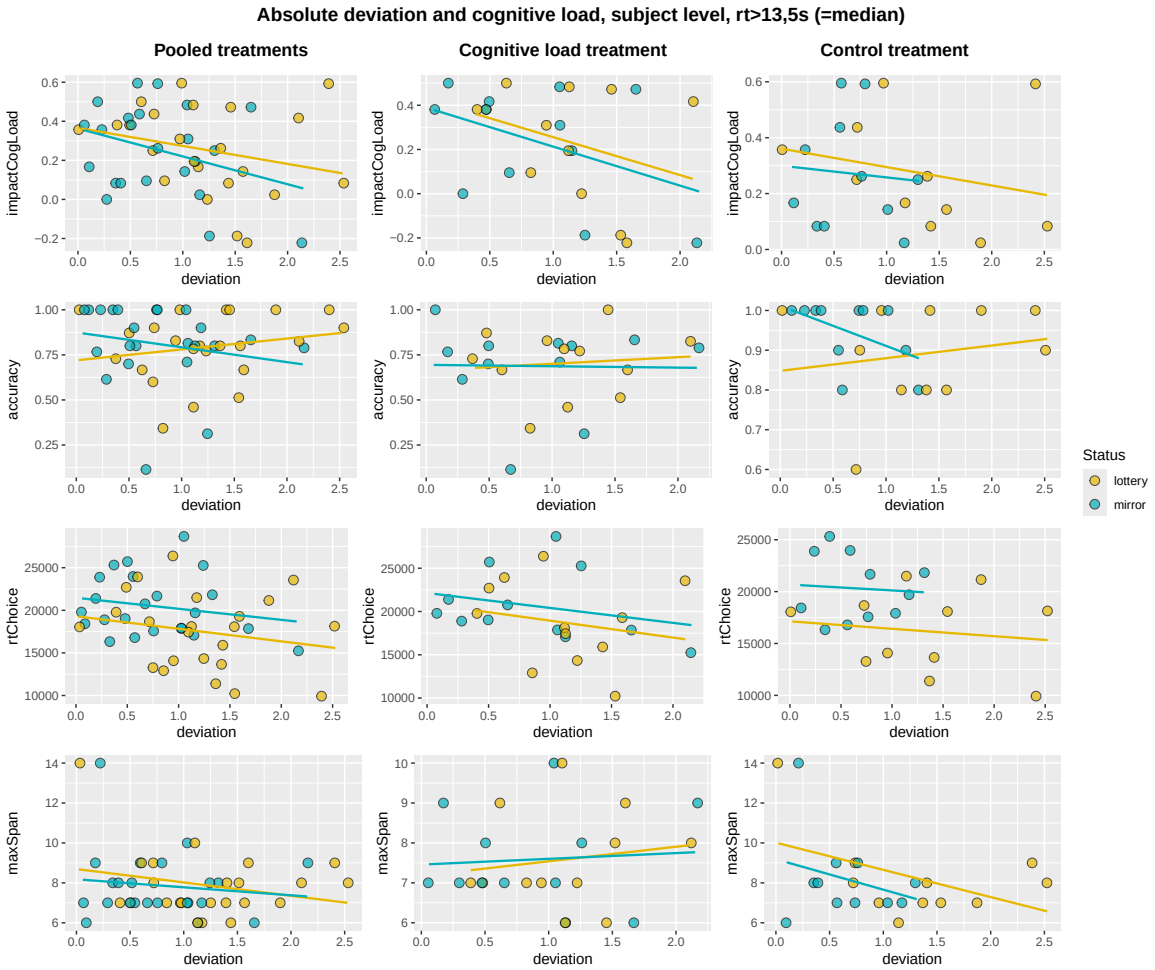

Figure 6: In this figure, all participants took on average more than 13,5 seconds to make their decisions on the choice tasks. The x-axis represents the mean absolute deviation for each subject in the lottery (yellow) and mirror (blue) tasks. On the first row the y-axis represents the difference between the source task and the target tasks in the first part of the experiment. A very cognitive load sensitive subject would have a high positive difference. The y-axis of the second row represents the mean accuracy on the cognitive load task (span task) for the lotteries and mirrors tasks. On the third row the y-axis indicates the mean time that each participants has spent on the task (the MPLs). The last row's y-axis represents the maximum span of each subject. The lines are linear regression fits of the two variables, for lotteries and mirror choices. In total 45 subjects passed the comprehension questions, 23 are in the cognitive load treatment and 22 in the control treatment.

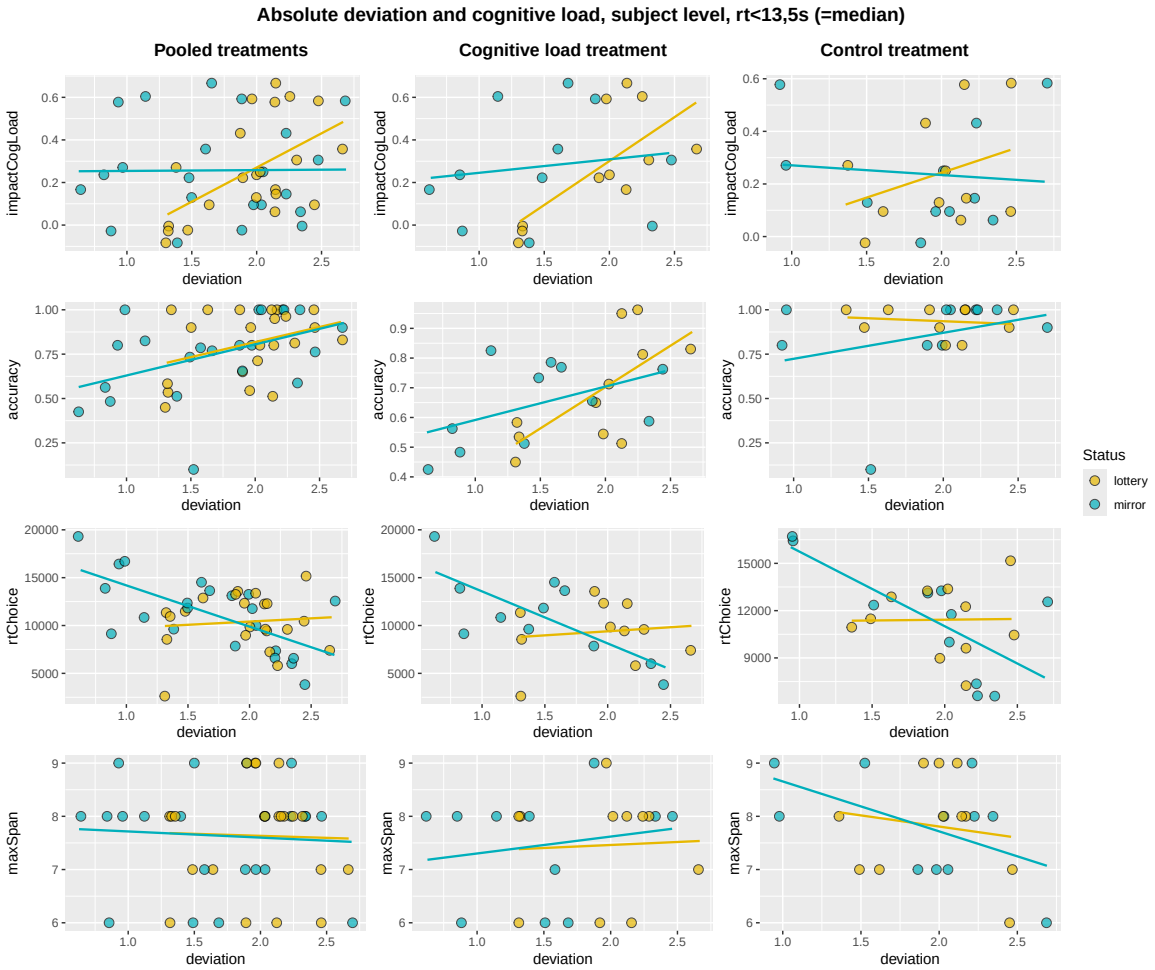

Figure 7: In this figure, all participants took on average less than 13,5 seconds to make their decisions on the choice tasks. The x-axis represents the mean absolute deviation for each subject in the lottery (yellow) and mirror (blue) tasks. On the first row the y-axis represents the difference between the source task and the target tasks in the first part of the experiment. A very cognitive load sensitive subject would have a high positive difference. The y-axis of the second row represents the mean accuracy on the cognitive load task (span task) for the lotteries and mirrors tasks. On the third row the y-axis indicates the mean time that each participants has spent on the task (the MPLs). The last row's y-axis represents the maximum span of each subject. The lines are linear regression fits of the two variables, for lotteries and mirror choices. In total 45 subjects passed the comprehension questions, 23 are in the cognitive load treatment and 22 in the control treatment.

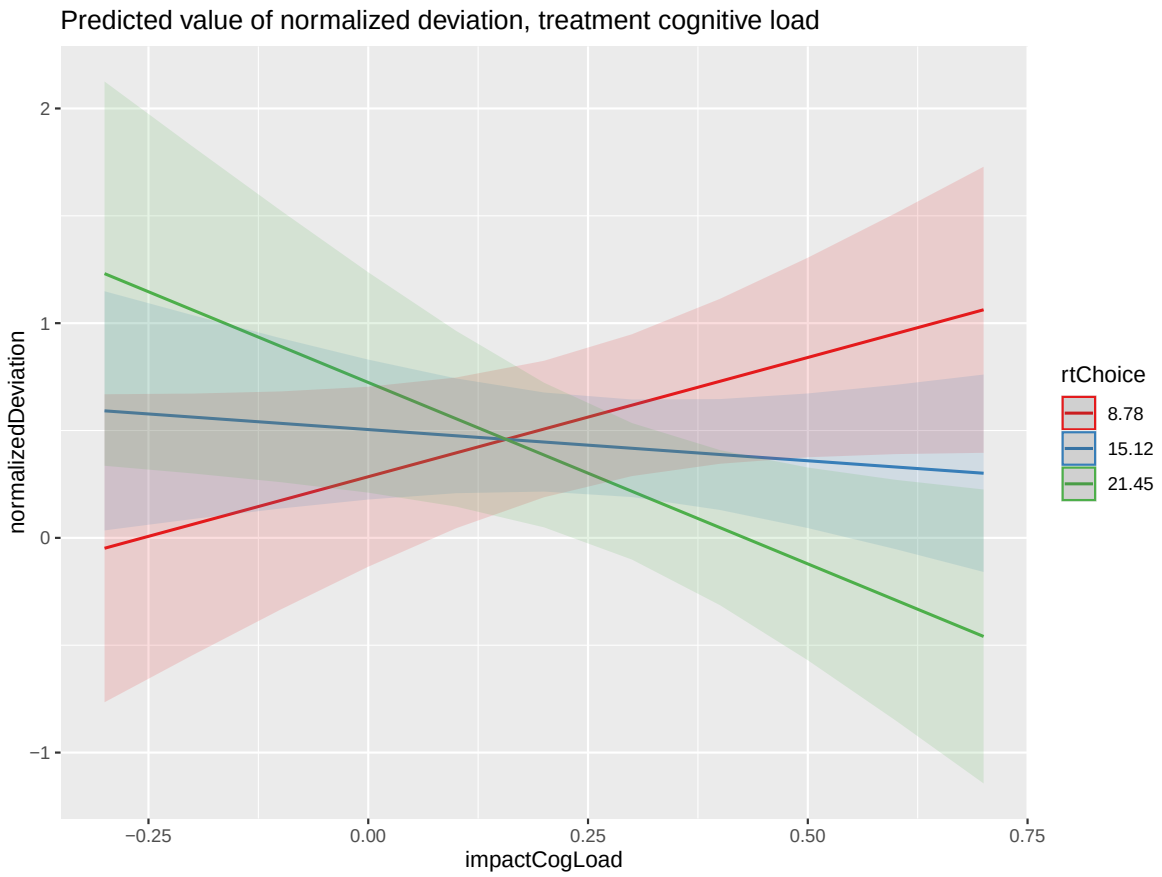

Figure 8: The figure shows the prediction of the model with sensitivity to cognitive load and time spent on choices, for three levels of time spent on the MPLs, in the cognitive load treatment. Mean sensitivity to cognitive load in the cognitive load treatment: 0.254.

3. There is a negative correlation between time and cognitive load sensitivity for both absolute and normalized deviations, which could explain why the cognitive load does not lead to a between-treatments difference.

We can see in figure 8 that sensitivity to cognitive load interacts strongly with the time spent on choices.

2.3.3 Deviation difference between treatments

To confirm that the cognitive load treatment has no impact (path (i)), I fit another model, specified as following. I regress the deviation for each subject i and task type (e.g. G10) q and task status (lottery or mirror) s on the mean time taken RT and the subject's treatment COG.

$$dev_{iqs} = \alpha + \beta \cdot (\text{RT})_{iqs} + \gamma \cdot (\text{COG})_i + \varepsilon_{iqs}$$

We can see in the regression Table 2 that the time spent seems to affect the noise but not in the direction of PT: the effect is significant only on absolute deviations. Cognitive load has no significant effect overall, probably because of the interaction with sensitivity to cognitive load as shown in Table 1 and no interaction is found. This suggests that given the sensitivity to cognitive load of the participants, lowering the time pressure would not result in more deviations.

3 Conclusion

There is an effect of cognitive load: mediated by response time, cognitive load sensitivity and accuracy correlates with the deviations. But this effect appears too small to significantly alter deviations.

	(1)	(2)
cognitive load sensitivity (subject level)	3.052**	1.828*
	[0.876, 5.227]	[0.396, 3.260]
Time on MPLs	0.035	−0.037*
	[−0.013, 0.082]	[−0.072, −0.002]
interaction	−0.221**	−0.113*
	[−0.355, −0.087]	[−0.211, −0.014]
Num.Obs.	46	46
R2	0.189	0.436

Table 1: Impact of cognitive load and time spent on normalized deviations in the cognitive load treatment. In square brackets are the 95% confidence intervals, calculated with SE clustered at the participant level (see Figure 10).

	(1)	(2)
time on MPLs	−0.04	−0.10***
	(0.03)	(0.01)
cognitive load treatment	−0.38	−0.67
	(0.63)	(0.33)
Interaction	0.02	0.04
	(0.04)	(0.02)
R^2	0.05	0.42
nobs	90	90

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 2: Effect of the time spent ont he MPLs on normalized (left panel) and absolute (right panel) deviation, depending of the treatment. Robust SE in parentheses, clustered at participant level.

Moreover, reducing the time spent on tasks would also increase deviations and normalized deviations in the no-cognitive-load treatment (see Figure 9. Participants may not be sensitive to the cognitive load enough.

References

[Choi et al., 2022] Choi, S., Kim, J., Lee, E., and Lee, J. (2022). Probability weighting and cognitive ability. *Management Science*, 68(7):5201–5215.

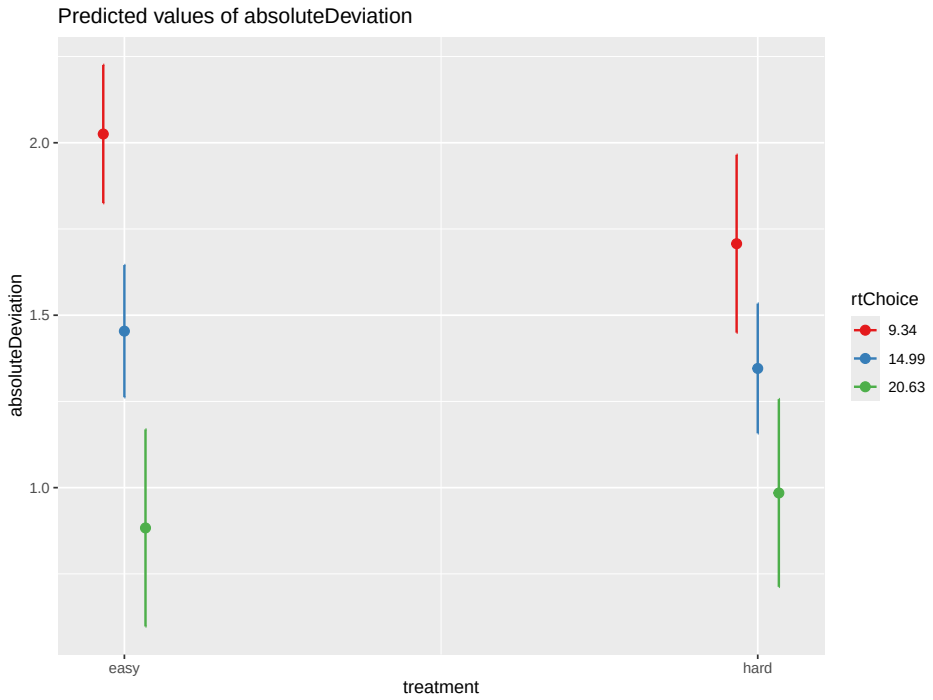

Figure 9: This figures plots the prediction of the second linear model presented above, regressing absolute deviations on the the treatment and the time spent on the tasks (MPLs). Error bars represent 95% confidence interval (with robust standard errors, clustered at the individual level).

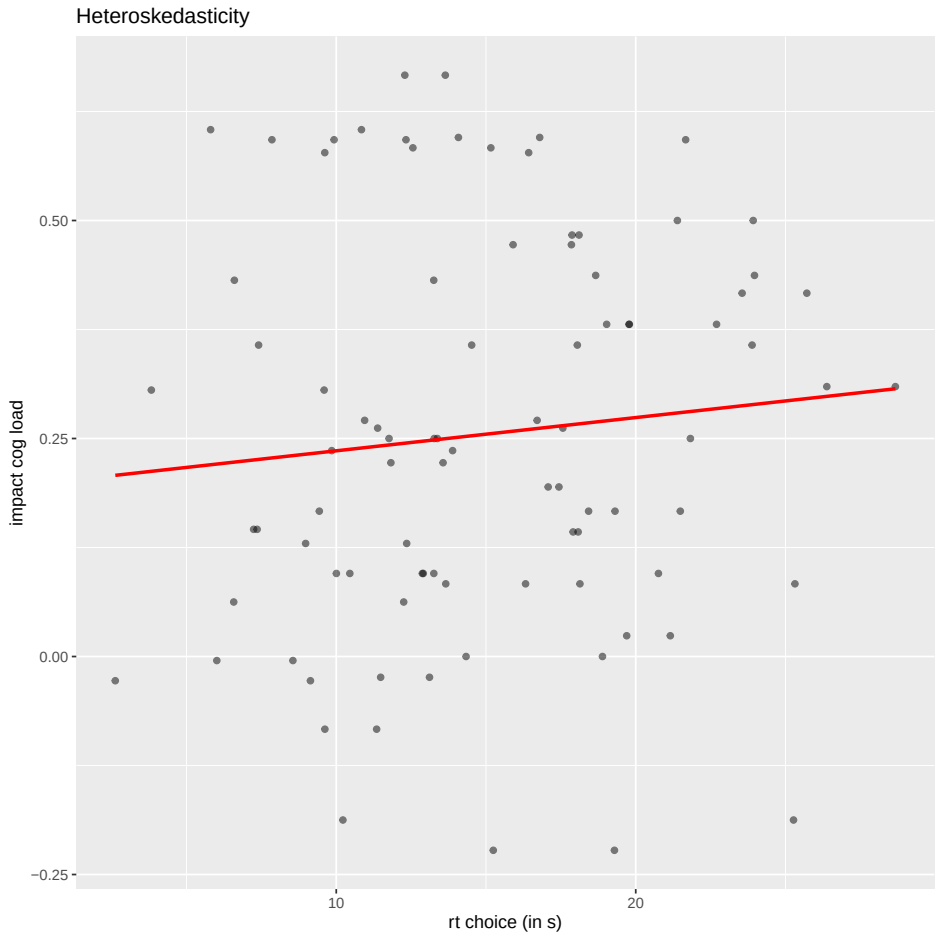

Figure 10: This figures plot the impact of cognitive load at the individual level on the reaction time for lotteries and for mirrors. The spread of the data doesn't seem to be correlated with either of the variables.